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Spreadsheets and simple databases are the most widely used data systems in the world. A shop owner keeps sales and inventory in Excel; a teacher keeps grades and attendance; a clinic keeps patient visits and medicine batches; an HR officer keeps timesheets; a cooperative keeps yields and input costs; a commune office keeps household statistics; a journalist downloads open data from the General Statistics Office; an employee of a large company wants a number from the data warehouse without waiting for the data team. All of them have the data. Almost none of them can write a query. What they want is to ask in plain Vietnamese — "Tháng này bán chạy nhất mặt hàng gì?", "Tuần rồi bán được bao nhiêu ly trà sữa?", "Còn ai nợ tiền chưa trả?", "Thuốc nào sắp hết hạn trong kho?" — and get a correct answer.

DataMate challenges you to build an AI agent that answers natural-language Vietnamese questions directly on structured data (CSV, Excel, SQLite): given the raw files and a question, the agent returns a correct answer — or **declines to answer** when the data cannot support the question. The competition data is anchored in the world of Vietnamese small businesses — the largest group of spreadsheet users in the country and the one with the richest, messiest real-world data — but the capability you build is general: natural-language question answering over tabular data.

Input:

- **A data package** for one business: one or more files (.csv, .xlsx, .sqlite) plus a short README.md describing the business.
- **A question** in everyday Vietnamese.
- **An as-of date**: the "today" against which relative time expressions ("tháng này", "tuần rồi", "cùng kỳ năm ngoái") are resolved.
- **A rulebook** and a **solar–lunar calendar**, shared by every business of the split (see the Data section)).

Output:

- A **final answer** of one of the types in the Evaluation section: a value (number, percent, string, date, boolean, list, table), unanswerable when the data cannot support the question, or abstain when the agent chooses not to answer.

An example:

Input — one line of questions.jsonl:

```
{"id": "q_06de33fa21cc", "business_id": "biz_118", "question": "sp bán dc nhiều nhất tuần rồi là gì", "as_of_date": "2026-03-04"}
```

Output — one line of submission.jsonl:

```
{"id": "q_06de33fa21cc", "type": "string", "answer": "Xi măng Hà Tiên PCB40 bao 50kg"}
```

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Competition rounds

The competition runs on a leaderboard with two rounds:

- **Round 1 — Public test:** teams develop their solutions, run them on the public test set and upload submission.jsonl. Scores appear on the public leaderboard and are for reference only during development.
- **Final solution submission:** before the solution deadline, the final solution is submitted as a Docker image, following the organizers' separate Docker guideline. For scoring, the organizers mount the test set into the container at /data and collect /result/submission.jsonl.
- **Round 2 — Private test:** after teams submit and freeze their solutions, the organizers release the private test set. Teams run their frozen solution on it and upload submission.jsonl. The final ranking is based on the organizers' re-run of the submitted Docker images (Rule 8).

Technical requirements

Every solution must meet all of the following requirements:

1. **Teams and eligibility** follow the general regulations of Zalo AI Challenge 2026.
2. **Models.** You may use any model whose weights were publicly released before 1 October 2026 under a licence that permits such use, and you may fine-tune it freely. Across all models used at inference by your solution (language models, embedding models, rerankers, classifiers), whether loaded at the same time or not, the **total number of parameters must not exceed 15 billion**, counting all parameters of mixture-of-experts models. Any quantization is allowed. The models used must be declared in the solution README.
3. **No external services.** Inference must run fully offline inside your Docker image: no calls to commercial LLM APIs, no downloads, no network access of any kind. The image is run with networking disabled.
4. **No human in the loop.** Answers must be produced by your system without manual intervention. Hard-coding answers, or logic keyed on question or business identifiers, is prohibited.
5. **Code execution.** Your agent may generate and execute code inside the container; this is expected but not required. The data directory is mounted read-only; write temporary files elsewhere.
6. **Data.** You may use the provided training data, data you synthesize yourself, and publicly available datasets and models. You may not use the public or private test data for training, nor collect answers to test questions by any means.
7. **Compute reference.** Solutions are verified on a machine with **2 × NVIDIA RTX 5090 (32 GB VRAM each, 64 GB total), 512 GB RAM**. Your solution must process the entire private test set within **6 hours** of wall-clock time on this machine, including model loading. Blackwell GPUs require CUDA ≥ 12.8 and a matching PyTorch/vLLM build; the Docker guideline specifies the base image to use.
8. **Final score and reproducibility.** The score of the organizers' re-run of your submitted Docker image on the private test set is your final score. The submission.jsonl you upload during the private test phase determines only the provisional leaderboard and which teams are re-run. Your solution must be deterministic enough that the two runs agree: a solution is considered reproducible if $|PA(\text{uploaded}) - PA(\text{re-run})| \leq 0.03$. If the difference exceeds 0.03, the organizers will examine the logs written by your image (see Submission format, Section Data), may run the image once more, and may request additional information from the team; if the difference cannot be explained, the entry is disqualified. If the re-run exceeds the time limit of Rule 7 or terminates early, only the answers written to /result/submission.jsonl before termination are scored; the rest count as abstain.

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Per-question scoring

Each answer is scored **correct**, **wrong** or **abstain**. An answer of type abstain (or a missing answer) is an abstain. Any other answer is correct only if its type equals the reference type **and** its value matches the reference as follows; otherwise it is wrong. No partial credit.

Type	Answer content	Example	Matching rule
`number`	A numeric value (VND amounts, counts, quantities)	`48250000`	Exact match. Reference values are integers (VND amounts, counts, quantities) or rounded to 2 decimals.
`percent`	A percentage value, in percent	`-12.45`	Exact match after rounding to 2 decimal places; the sign must be correct.
`string`	A name (product, customer, branch, weekday, …)	`"Sting dâu 330ml"`	Both the answer and each accepted form of the reference are normalized, then compared for exact equality. Normalization: (1) Unicode NFC; (2) lowercase; (3) every character that is not a letter, a digit or whitespace is replaced by a space; (4) leading/trailing whitespace removed and runs of whitespace collapsed to one space. Diacritics are not removed. Each reference lists its accepted forms: every spelling of the name that occurs in that business's data, plus its code — after normalization, e.g. `sting dâu 330ml`, `sp0042`.
`date`	A date in ISO format	`"2026-08-15"`, `"2025-09"`	Exact match. `yyyy-mm-dd`; questions about a month: `yyyy-mm` (e.g. `"2025-09"`).
`boolean`	`true` / `false`	`true`	Exact match.
`list`	An ordered list of names or numbers, in the order the question asks for (or the rulebook default)	`["Cà phê sữa", "Bạc xỉu", "Trà đào"]`	Always ordered, and matched position by position. The order is the one the question asks for, or the rulebook default when the question gives none (see Ranking questions below). Elements are matched with the `string` or `number` rule.
`table`	A small table: column names and rows	`{"columns": [...], "rows": [[...], ...]}`	Columns are matched by normalized name; if the names do not match, by position. Rows are matched as a set; name cells with the `string` rule, numeric cells with the `number` rule.
`unanswerable`	The data does not contain what is needed to answer	—	Correct if and only if the reference is `unanswerable`.
`abstain`	The agent chooses not to answer	—	Scored as abstain regardless of the reference.

Ranking questions. Rankings use the criterion stated in the question, or the rulebook default when the question gives none, in descending order of that criterion over valid orders only; elements with equal values are ordered by code, ascending. Questions whose reference ranking has a tie at the k boundary (or at rank 1 for "… nhất" questions) are excluded from the test sets. When fewer than k candidates exist, the reference answer is all of them.

Score

The leaderboard metric is **Penalized Accuracy (PA)** over the N questions of a test set:

$$\mathrm{PA} = \frac{1}{N}\sum_{i=1}^{N} s_i, \qquad s_i = \begin{cases} 1 & \text{answer } i \text{ is correct} \\ 0 & \text{answer } i \text{ is } \texttt{abstain} \\ -1 & \text{answer } i \text{ is wrong} \end{cases}$$

Any answer given to an unanswerable question counts as wrong. All questions carry equal weight. Ties in the final ranking are broken by total inference time in the organizers' re-run, then by earlier submission.

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Business types

The data covers a wide range of Vietnamese small businesses: grocery stores, coffee and bubble-tea shops, restaurants, pharmacies, building-material stores, motorbike repair shops, mother-and-baby stores, and more. The private test includes business types that do not appear in the training or public test data.

Each business covers 12–30 months of activity with seasonality (Tết, Mid-Autumn, public holidays, paydays, weekends) and ranges from a few hundred to about 300,000 rows.

Package structure

```
<split>/
├── data/<business_id>/
│   ├── *.csv | *.xlsx | *.sqlite     # one or more data files
│   └── README.md                     # business description and notes
├── rulebook.md                       # default rules and vocabulary
├── calendar.csv                      # solar–lunar calendar (Tết)
├── questions.jsonl                   # one question per line
└── answers.jsonl                     # training only
```

Sample record in questions.jsonl:

```
{"id": "q_01234", "business_id": "biz_017", "question": "thang nay hang nao ban cham nhat, ton nhieu ma ban ko dc", "as_of_date": "2026-08-27"}
```

Training businesses also come with answers.jsonl, which contains the reference answer for every training question, linked by "id". Each line follows the submission format, plus an "alias" field listing other accepted forms of a name answer (alternative spellings found in the data and the item code). Unanswerable questions have "answer": null. Submissions do not need to include the "alias" field.

Sample record in answers.jsonl (the reference answer of the example question in the Description section):

```
{"id": "q_06de33fa21cc", "type": "string", "answer": "Xi măng Hà Tiên PCB40 bao 50kg", "alias": ["SP0001"]}
```

The quality of README.md varies: some businesses come with a full data dictionary, others with a few informal notes, some with almost nothing.

Splits

Split	Businesses	Questions	Reference answers
Training	30	~3,000	Yes
Public test	10–20	~1,000	Hidden; scored on the public leaderboard
Private test	10–20	~500	Hidden; released at the private test phase; includes new business types

Business rules (default interpretations)

Each split ships its own rulebook (`rulebook.md`), which fixes the default meaning of common terms whenever the question does not say otherwise. The rulebook may differ between splits so your agent must read and apply the rulebook of the split it is answering, not a copy of the training rulebook. Typical rules: revenue = sum of line amounts after

line-level discounts (order-level vouchers are not deducted), excluding VAT and delivery fees, excluding cancelled orders and orders with any returned item; "bán chạy" = by quantity; a week runs Monday–Sunday; "tháng này" = the month containing the as-of date, up to and including that date; default cost = weighted-average purchase cost by lot; the Tết window = 23 tháng Chạp to mùng 10 (lunar calendar).

Submission format

Submission file. In both rounds you upload a submission.jsonl file, one JSON object per line:

```
{"id": "q_01234", "type": "string",  "answer": "Sting dâu 330ml"}
{"id": "q_01235", "type": "number",  "answer": 48250000}
{"id": "q_01236", "type": "percent", "answer": -12.45}
{"id": "q_01237", "type": "date",    "answer": "2026-08-15"}
{"id": "q_01238", "type": "boolean", "answer": true}
{"id": "q_01239", "type": "list",    "answer": ["Cà phê sữa", "Bạc xỉu", "Trà đào"]}
{"id": "q_01240", "type": "table",   "answer": {"columns": ["khach_hang", "so_tien_no"], "rows": [["Chị Lan", 3200000], ["Anh Tú", 1500000]]}}
{"id": "q_01241", "type": "unanswerable", "reason": "Dữ liệu không có giá vốn nên không tính được lợi nhuận"}
{"id": "q_01242", "type": "abstain"}
```

Questions missing from the file are scored as abstain. The reason field is optional. The number of submissions per day follows the general regulations.

Docker image. The image (see Competition rounds in the Rule section) must contain the source code (training and inference), the model weights and a predict.sh that reads the test set from /data and writes /result/submission.jsonl, together with a README. It should also write a per-question log to /result/logs/, in any format, containing at least the question id, the final answer and the elapsed time — plus anything else that shows how the answer was reached (e.g. generated code, execution output). The organizers use these logs to resolve differences between your uploaded result and their re-run (Rule 8).

Data Train: Download
Public Test: Download

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I. Public Test Round (October 26 – November 16)

- Participating teams build their own solutions, run them on the organizers' public test set, and submit result files (JSONL) via the Submission section on the Zalo AI Challenge portal. Teams are allowed up to 5 submissions per day, and the quota will be reset at the end of each day. Scores and rankings will be displayed on the public leaderboard.
- On November 16, participating teams must submit their complete solutions, including training code, inference code, and data, in the form of a Docker package to the organizers via the "Solution Submission" form on the Zalo AI Challenge portal. Since the Docker file is usually quite large, teams should provide a checksum and a public Google Drive link so that the organizer can download it.

Each team is allowed up to 2 submissions. The organizer will only evaluate the last submission. Submission deadline: 23:59 (GMT+7).

II. Private Test Round (November 20 – November 23)

- At 9:00 AM on November 20, the organizer will release the private test set. Participating teams will run their submitted solutions on this set and submit the result files (JSONL) via the Submission section on the Zalo AI Challenge portal, with a deadline of 23:59 (GMT+7) on November 23. The private leaderboard will be displayed in chronological order (based on submission time).

The organizer will evaluate the submitted solutions to verify results, review source code, and detect any cheating. The top 2 teams will be announced on December 2.